Exp 6: Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging in Zoho Creator

1. Aim

To automate the gym enquiry process in Zoho Creator by configuring a form workflow that runs when an enquiry is submitted, writing Deluge scripts that check the entered age and display the captured values for debugging, and sending an automatic confirmation email to the person who submitted the enquiry, then testing the workflow with sample submissions.

2. Procedure

2.1 Automation Requirements

Without automation, the front desk has to read every new enquiry, check the details and reply by hand. Four steps were automated with one workflow and its Deluge script, as listed in Table 1.

Requirement	Component	Details
Run logic after an enquiry is saved	Form workflow “Form submission”	Record event: Created. Form event: Successful form submission
Check that the age entered is sensible	Deluge if-else condition	Customer_Age must lie between 0 and 100
See the captured values while testing	Deluge info statements	Customer_Name, Email_Address, Age and Preferred_Slot
Acknowledge the enquiry to the customer	Deluge sendmail task	Email sent to the Email_Address entered in the form

Table 1: Automation requirements and the components used

2.2 Creating the Form Workflow

The workflow was created from **Workflows > Form Workflows** in the Creator builder, using the settings below (Figure 1):

- **Choose form:** the enquiry form of the application (shown as *Admission Enquiry Form*).
- **Run when a record is:** *Created*, so the workflow fires only for new enquiries.
- **When to trigger workflow (form event):** *Successful form submission* (On success). This event runs after the record has been saved, which makes it the right place for emails and other follow-up actions.
- **Name the workflow:** *Form submission*, followed by **Create Workflow**.

A Deluge script action was then added on the workflow canvas, and all the scripts below were written inside it.

2.3 Deluge Script: Age Check

The first block uses an if-else condition to test whether the submitted age is a realistic value. A value from 0 to 100 produces a success message; anything else produces an error message. The check reports its result through **info** statements and does not cancel the submission, because the blocking validation (age 14 to 75) already exists in the On validate script built in Exp 4 (Figure 2).

```
if(input.Customer_Age >= 0 && input.Customer_Age <= 100)
{
    info "Age validation successful";
}
else
{
    info "Invalid age";
}
```

Script 1: Age validation using an if-else condition

2.4 Testing and Debugging with info Statements

To confirm that the form was passing the right values into the script, **info** statements were added that print the customer's name, email address, age and preferred slot (Figure 3). The info statements write to the application's execution log, which makes it possible to see what the script received without changing any data. This is the main debugging aid in Deluge, because Creator has no step-through debugger.

```
info "Customer Name: " + input.Customer_Name;
info "Email: " + input.Email_Address;
info "Age: " + input.Age;
info "Preferred Slot: " + input.Preferred_Slot;
```

Script 2: info statements used to debug the values captured by the form

The workflow was tested by submitting sample enquiries in the live application and checking the log and the mailbox. Table 2 lists the cases covered and the behaviour the script is designed to give.

Test case	Input	Expected behaviour
Valid age	Age within 0 to 100	Log shows "Age validation successful"
Invalid age	Age below 0 or above 100	Log shows "Invalid age"
Captured values	Any valid submission	Log lists the name, email, age and preferred slot entered
Confirmation email	Valid email address entered	Customer receives the confirmation email (Figure 4)

Table 2: Test cases for the form submission workflow

2.5 Deluge Script: Confirmation Email

The last action sends an acknowledgement to the address the customer typed into the form, using the Deluge **sendmail** task. The sender is the application's default notification address, so no mail configuration was needed.

```
sendmail
[
    from: zoho.adminuserid
    to: input.Email_Address
    subject: "Gym Enquiry Submitted Successfully"
    message: "Dear Customer,<br><br>" +
             "Your gym enquiry has been successfully submitted.<br><br>" +
             "Our Gym Team will review your enquiry and contact you<br>" +
             "regarding gym membership and available slots.<br><br>" +
             "Thank you for your interest.<br><br>Regards,<br>Gym Team"
]
```

Script 3: sendmail task for the confirmation email

3. Output

The following figures show the workflow settings, the Deluge scripts and the confirmation email received after a test submission.

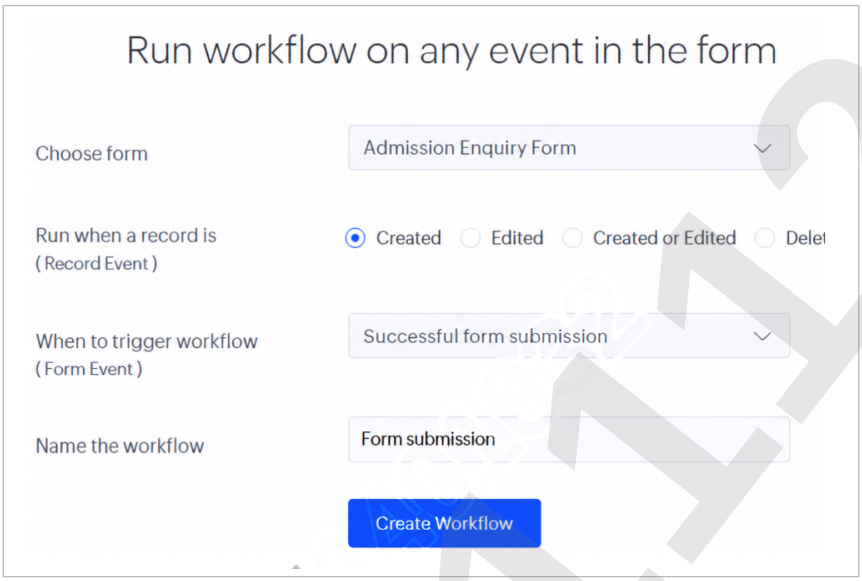


Figure 1: Form workflow settings (Created, Successful form submission, named “Form submission”)

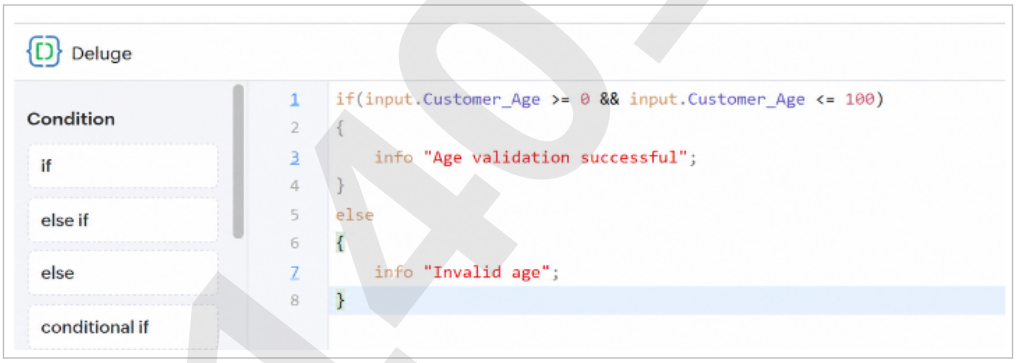


Figure 2: Deluge script for the age check

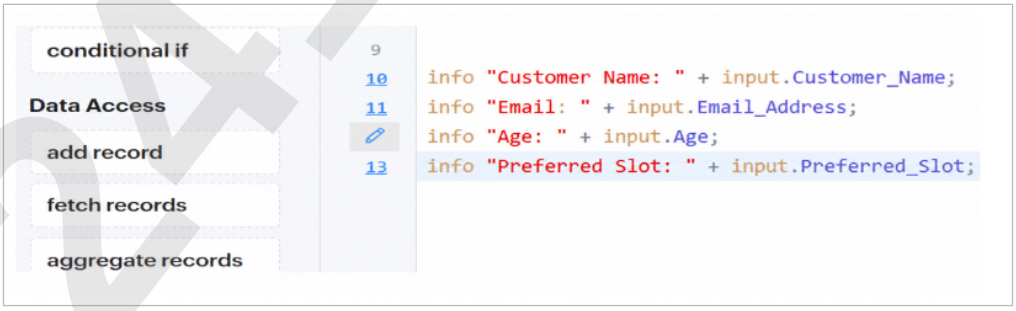


Figure 3: info statements used to display the captured form values

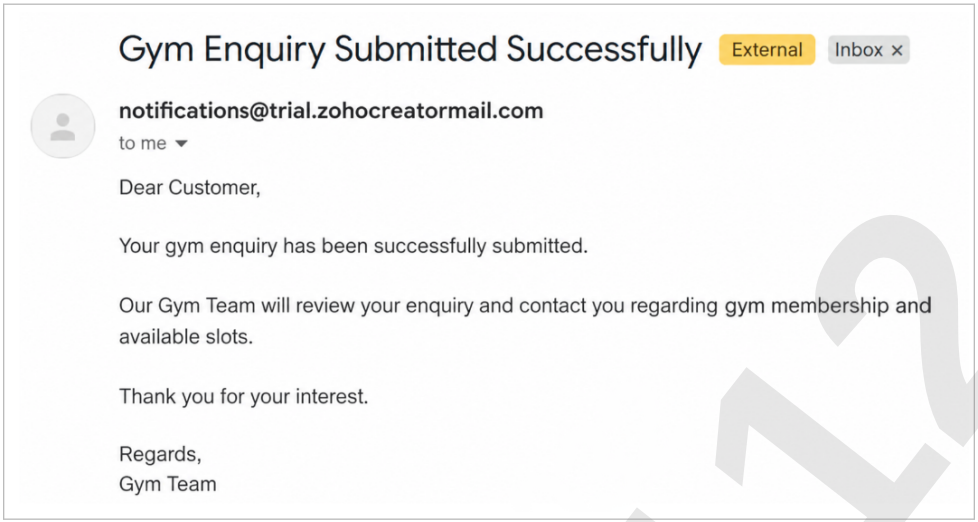


Figure 4: Confirmation email "Gym Enquiry Submitted Successfully" received in the mailbox

Observations from the output

- The workflow runs only on the Created event and only after the record is saved, so the customer is never emailed for a submission that failed.
- The confirmation email arrived from the Creator notification address and carried the subject and message written in the script.
- The info statements make the script's input visible, which is the quickest way to trace a wrong value back to the field it came from.
- Keeping the blocking age rule in the On validate event and the reporting check in On success separates input validation from post-submission automation.

4. Result

A form workflow was successfully created in Zoho Creator that runs on successful submission of a new gym enquiry. Deluge scripts checked the age, displayed the captured values for debugging, and sent an automatic confirmation email to the customer, which was verified with test submissions. This completes the automation, testing and debugging requirements of this experiment.

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